MMRM vs. Survival Analysis for AD Clinical Trials: A Historical Review and Simulation Study

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Contents

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1 Introduction

1.1 The divergent paths of clinical trial analysis

The statistical analysis of clinical trials has followed markedly different trajectories across therapeutic areas. In oncology, time-to-event endpoints analyzed by survival methods remain the regulatory gold standard, with overall survival serving as the definitive measure of treatment efficacy[1,2]. In contrast, Alzheimer's disease (AD) and other neurodegenerative disease trials have converged on change-from-baseline estimands analyzed by the mixed model for repeated measures (MMRM) as the primary analytic framework[3,4]. This divergence is not an accident of history but rather reflects fundamental differences in disease biology, regulatory philosophy, and the statistical properties of the available endpoints.

We trace the historical developments that led to the current analytic conventions in each field and present a simulation study comparing MMRM and survival analysis applied to the same underlying disease process, using the Clinical Dementia Rating (CDR) scale as the primary measure.

1.2 Regulatory origins: the FDA dual-outcome requirement

The regulatory framework for AD clinical trials was shaped decisively by the 1990 FDA guidelines authored by Paul Leber, which required demonstration of efficacy on two co-primary endpoints: one cognitive (such as the Alzheimer's Disease Assessment Scale-Cognitive Subscale, ADAS-Cog) and one global or functional measure[5,6]. The rationale was that a statistically significant difference on a psychometric test alone might not reflect clinically meaningful change; a second, independent measure of global function was needed to confirm relevance. This dual-outcome requirement established the precedent that AD trials would be evaluated on the magnitude and direction of change from baseline, rather than on the occurrence of discrete clinical events.

The ADAS-Cog, developed by[6], and the CDR, codified by[7], became the two pillars of AD trial assessment. The CDR in particular provided both a global staging instrument (the global CDR score, ranging from 0 to 3) and a sum-of-boxes score (CDR-SB, ranging from 0 to 18) that offered a quasi-continuous measure of disease severity suitable for change-score analysis. The 2018 FDA guidance on early AD further reinforced the change-from-baseline paradigm, while allowing greater flexibility in endpoint selection for earlier disease stages[8].

1.3 The fall of LOCF and the rise of MMRM

Through the 1990s, the standard analytic approach for AD trials was last observation carried forward (LOCF) combined with analysis of covariance (ANCOVA) at the final time point. Under LOCF, a participant's last observed value is imputed for all subsequent missing visits, and the analysis proceeds as if no data were missing. This approach was computationally convenient and widely accepted by regulators, but it rests on the untenable assumption that a participant's trajectory would have remained flat from the time of dropout (an assumption grossly violated in progressive diseases where dropout is often driven by worsening symptoms).

The foundational work of[9] and[10] demonstrated through simulation that LOCF produces substantially biased treatment effect estimates and inflated Type I error rates when dropout is differential between treatment arms. In a progressive disease like AD, where placebo-arm participants tend to drop out at higher rates due to worsening, LOCF paradoxically imputes favorable values for the placebo group (freezing them at an earlier, less impaired state), thereby attenuating the estimated treatment effect. Conversely, if active-treatment participants drop out due to adverse events while still relatively well, LOCF preserves their favorable scores, inflating the apparent benefit.

[4] and[11] extended this work by comparing LOCF to the mixed model for repeated measures (MMRM), which uses all observed data under a likelihood-based framework that is valid when data are missing at random (MAR). Unlike LOCF, MMRM does not impute missing values; instead, it borrows information from the observed correlation structure across time points to provide unbiased estimates under MAR.[12] proposed alternative semi-parametric and non-parametric methods and found that MMRM performed comparably to their proposed approaches while making less restrictive assumptions.[13] provided the theoretical underpinning for likelihood-based analyses of incomplete longitudinal data.

By the mid-2000s, the weight of evidence against LOCF was overwhelming, and the 2010 National Research Council panel report, commissioned by the FDA, formally recommended against single imputation methods and endorsed likelihood-based and multiple imputation approaches as preferred alternatives[14]. The ICH E9(R1) addendum on estimands further clarified the conceptual framework by requiring explicit specification of the target estimand (including how intercurrent events such as treatment discontinuation are handled) before selecting an analytic method[15].

1.4 MMRM as the primary analysis in AD

[3] compared MMRM, slope models, and quadratic models across five AD Cooperative Study (ADCS) trials and found that MMRM, which treats time as categorical and imposes no parametric assumptions on the mean trajectory, provided robust estimates across a range of disease stages and trial durations. Although continuous time (slope) models can be more statistically efficient when the linearity assumption holds, the unstructured mean of MMRM offers protection against model misspecification in a disease where trajectories are heterogeneous and potentially nonlinear.[16] proposed a disease progression model (DPM) with proportional treatment effects and demonstrated power gains over MMRM in autosomal dominant AD, though the DPM requires stronger parametric assumptions that may limit generalizability.

Today, virtually every pivotal AD trial uses MMRM or a close variant as the pre-specified primary analysis. The aducanumab EMERGE and ENGAGE trials[17] and the lecanemab Study 201[18] all employed MMRM with change from baseline in CDR-SB as the primary endpoint, with sensitivity analyses from DPM and other longitudinal models used to assess robustness.

1.5 Why survival analysis persists in oncology

In oncology, the natural endpoint is death, and overall survival (OS) has served as the primary measure of treatment efficacy since the earliest cancer clinical trials. OS is binary, unambiguous, and universally meaningful.[2] cautioned against replacing OS with surrogate endpoints unless rigorous validation criteria are met, and the FDA has historically required OS data for full (as opposed to accelerated) approval of cancer drugs[1]. Even when progression-free survival or response rate serves as the primary endpoint for accelerated approval, confirmatory trials typically require OS data.

The persistence of survival analysis in oncology reflects several features that distinguish it from AD: (1) a clearly defined, irreversible event (death); (2) a well-understood proportional hazards framework with established regulatory precedent; (3) a tradition of interim analyses based on information fractions from event counts; and (4) the absence of a continuous, validated severity measure that changes monotonically and is universally accepted across tumor types.

1.6 Rate of change vs. time to event in AD

Given that CDR and similar scales can be used to define both change scores (amenable to MMRM) and threshold crossings (amenable to survival analysis), the question arises: what is the relative efficiency of these two approaches when applied to the same underlying data?

[19] derived an analytic inflation factor comparing Cox proportional hazards models of time to a threshold with linear models of continuous outcomes, and demonstrated through simulation that linear models consistently yielded greater statistical power for detecting treatment effects in ADNI data.[20] extended this work to presymptomatic AD, finding that power was approximately doubled with repeated continuous measures compared to time-to-progression analysis.[21] introduced a progression model for repeated measures (PMRM) that expresses treatment effects in time units (days of delay) rather than CDR-SB points, offering a clinically interpretable alternative while maintaining efficiency comparable to MMRM.[22] modeled CDR-SB trajectories to predict time to clinically meaningful worsening in MCI patients, finding that the median time to a 1-point change was approximately 2 years—informing both trial duration and sample size planning.

1.7 Present study

Despite the theoretical and empirical evidence favoring continuous measures, the comparison between MMRM and survival analysis for AD endpoints has not been systematically examined in a simulation that uses the global CDR as both a continuous outcome and the basis for a survival endpoint defined by conversion from CDR 0.5 (MCI/very mild dementia) to CDR 1.0 or higher (mild dementia). We address this gap through a Monte Carlo simulation that generates realistic CDR trajectories under a plausible data-generating model and compares the statistical properties of MMRM and Cox regression applied to the same simulated trial data.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther[23], the simulation is reported under the ADEMP framework.

- **Aims.** Compare the operating characteristics of an MMRM on change-from-baseline in latent CDR versus a Cox proportional hazards model on time to first CDR ≥ 1.0, under an Alzheimer's-disease DGM with informative dropout.
- **Data-generating mechanism.** Mixed-effects model for latent CDR trajectories with a discretized observed CDR and outcome-dependent dropout; details below.
- **Estimands.**
  (i) For MMRM: the treatment effect on change from baseline in latent CDR at the LAST visit, expressed as the linear combination $\beta_{\text{trt}} + \beta_{\text{trt:visit}_K}$.
  (ii) For Cox: the log-hazard ratio for time to first CDR ≥ 1.0.
- **Methods.** MMRM via `nlme::gls` with unstructured covariance; Cox via `survival::coxph`.
- **Performance measures.** Bias, empirical SE, mean model SE, power, coverage, and convergence rate, each reported with its Monte Carlo SE per Morris Table 6. With $n_{\text{sim}} = 2000$, the Monte Carlo SE for power at $p = 0.5$ is $\sqrt{0.5 \times 0.5/2000} = 0.011$; for coverage at 0.95, MCSE ≈ 0.005.

2.2 Data generating model

We simulate a two-arm, placebo-controlled trial with $n = 200$ participants per arm, all entering with a baseline global CDR of 0.5. The primary continuous outcome is the global CDR score observed at visits scheduled every 6 months over 24 months (5 post-baseline visits). The survival endpoint is time to conversion from CDR 0.5 to CDR ≥ 1.0.

The latent continuous disease severity Y_{ij}^* for participant i at visit j is modeled as:

$$Y_{ij}^* = \beta_0 + \beta_1 t_j + \beta_2 T_i + \beta_3 (t_j \times T_i) + b_{0i} + b_{1i} t_j + \epsilon_{ij}$$

where t_j is visit time in years, T_i is the treatment indicator (0 = placebo, 1 = active), b_{0i} and b_{1i} are participant-specific random intercepts and slopes drawn from a bivariate normal distribution, and $\epsilon_{ij} \sim N(0, \sigma^2)$ is residual error.

The observed global CDR is obtained by mapping Y^* to the CDR staging categories {0, 0.5, 1, 2, 3} using threshold parameters calibrated to observed CDR distributions in ADNI.

2.3 Parameter values

The simulation parameters are selected to reflect plausible progression rates in a population with MCI due to AD (CDR 0.5 at baseline):

Table 1: Simulation parameter values.

Parameter	Value	Description
β_0	0.5	Baseline CDR (MCI)
β_1	0.30	Annual rate, placebo (CDR/yr)
β_2	0.00	Baseline treatment effect
β_3	-0.075	Treatment x time (25% slowing)
σ_b^2 (intercept)	0.04	Random intercept variance
σ_b^2 (slope)	0.02	Random slope variance
ρ	0.3	Intercept-slope correlation
σ^2	0.01	Residual variance
Dropout rate	15%/year	MAR dropout, severity-dependent

2.4 Design specifications

- **Sample size:** 200 per arm (400 total)
- **Visit schedule:** Baseline, 6, 12, 18, 24 months
- **Survival endpoint:** Time from randomization to first visit with global CDR ≥ 1.0
- **MMRM endpoint:** Change from baseline in latent CDR score at 24 months
- **Treatment effect:** 25% slowing of progression (hazard ratio ≈ 0.75 for the survival endpoint)

2.5 Simulation procedure

```
sim_trial <- function(n_per_arm = 200,
                      n_visits = 5,
                      beta = c(0.5, 0.30, 0, -0.075),
                      sigma_b = matrix(
                        c(0.04, 0.3 * sqrt(0.04 * 0.02),
                          0.3 * sqrt(0.04 * 0.02), 0.02),
                        2, 2
                      ),
                      sigma_e = 0.1,
                      dropout_rate = 0.15) {
  n <- 2 * n_per_arm
  trt <- rep(0:1, each = n_per_arm)
  times <- seq(0.5, 2.5, by = 0.5)

  re <- MASS::mvrnorm(n, mu = c(0, 0), Sigma = sigma_b)

  records <- list()
  for (i in seq_len(n)) {
    y_star <- numeric(n_visits)
    dropped <- FALSE
    for (j in seq_len(n_visits)) {
      if (dropped) next
```

```r
      t_j <- times[j]
      mu_ij <- beta[1] + beta[2] * t_j +
        beta[3] * trt[i] + beta[4] * (t_j * trt[i]) +
        re[i, 1] + re[i, 2] * t_j
      y_star[j] <- mu_ij + rnorm(1, 0, sigma_e)
      cdr_obs <- cut(
        y_star[j],
        breaks = c(-Inf, 0.25, 0.75, 1.5, 2.5, Inf),
        labels = c(0, 0.5, 1, 2, 3)
      )
      cdr_obs <- as.numeric(as.character(cdr_obs))
      p_drop <- 1 - exp(-dropout_rate * 0.5 *
        (1 + 0.5 * (y_star[j] - 0.5)))
      if (runif(1) < p_drop) dropped <- TRUE
      records[[length(records) + 1]] <- data.frame(
        id = i, trt = trt[i], visit = j,
        time = t_j, y_star = y_star[j],
        cdr = cdr_obs
      )
    }
  }
  do.call(rbind, records)
}
```

For each of $R = 2000$ replications, we:

1. Generate a complete trial dataset using the model above.
2. Apply severity-dependent MAR dropout.
3. Fit an MMRM to the change from baseline in the latent CDR score, with treatment, visit, treatment-by-visit interaction, and baseline score as covariates, using an unstructured covariance matrix.
4. Fit a Cox proportional hazards model to time to first CDR ≥ 1.0, with treatment as the sole covariate.
5. Record the treatment effect estimate, standard error, p-value, and 95% confidence interval from each model.

2.6 Analysis plan

```r
# The estimand is the treatment effect at the LAST visit, expressed
# as beta_trt + beta_{trt:visit_f_last}. Earlier versions picked the
# `trt` main effect alone, which under the `trt * visit_f`
# parameterisation corresponds to the effect at the REFERENCE visit,
# not the final visit. See docs/morris-audit-2026-04-17.md.
fit_mmrm <- function(dat) {
```

```
  dat <- dat |>
    group_by(id) |>
    mutate(
      baseline = first(y_star),
      change = y_star - baseline
    ) |>
    ungroup() |>
    filter(visit > 0) |>
    mutate(visit_f = factor(visit))

  mod <- nlme::gls(
    change ~ trt * visit_f + baseline,
    data = dat,
    correlation = nlme::corSymm(form = ~ visit | id),
    weights = nlme::varIdent(form = ~ 1 | visit_f),
    na.action = na.omit
  )
  tt <- summary(mod)$tTable
  last_visit_level <- max(levels(dat$visit_f))
  L <- rep(0, nrow(tt))
  names(L) <- rownames(tt)
  L["trt"] <- 1
  L[paste0("trt:visit_f", last_visit_level)] <- 1
  est <- sum(L * tt[, "Value"])
  se <- sqrt(as.numeric(t(L) %*% vcov(mod) %*% L))
  p <- 2 * pnorm(-abs(est / se))
  c(est = est, se = se, p = p)
}

fit_cox <- function(dat) {
  surv_dat <- dat |>
    group_by(id) |>
    summarise(
      trt = first(trt),
      event_time = {
        conv <- which(cdr >= 1)
        if (length(conv) > 0) time[conv[1]] else max(time)
      },
      event = as.numeric(any(cdr >= 1)),
      .groups = "drop"
    )
  mod <- coxph(
    Surv(event_time, event) ~ trt,
```

```
    data = surv_dat
  )
  s <- summary(mod)
  c(
    est = coef(mod),
    se = s$coefficients[, "se(coef)"],
    p = s$coefficients[, "Pr(>|z|)"]
  )
}
```

2.7 Performance metrics

For each method, we compute:

- **Bias:** Mean estimate minus true parameter value
- **Empirical SE:** Standard deviation of estimates across replications
- **Mean model-based SE:** Mean of estimated standard errors
- **Power:** Proportion of replications with $p < 0.05$
- **Coverage:** Proportion of 95% CIs containing the true value
- **Convergence rate:** Proportion of replications producing valid estimates

3 Results

3.1 Headline findings

Under the data-generating model (true MMRM effect $= -0.15$, true log-hazard ratio $= \log 0.75 \approx -0.288$), the MMRM estimator recovered its estimand with bias -0.003 (MCSE 0.001), empirical SE 0.034, power 0.994, and coverage 0.954. The Cox model's log-hazard-ratio estimator had bias -0.002 (MCSE 0.002), empirical SE 0.110, power 0.716, and coverage 0.961. Convergence was 1.000 for MMRM and 1.000 for Cox across 2000 valid replications per method. Both estimands use the Morris Table 6 Monte Carlo SEs reported in Table 1 below.

3.2 Summary statistics

Table 2: Simulation results: MMRM vs Cox regression (2000 replications). Monte Carlo SEs per Morris et al. (2019) Table 6.

method	n_valid	bias	mcse_bias	emp_se	mcse_emp_se	mean_se	power	mcse_power	coverage	mcse_coverage	convergence
MMRM	2000	-0.003	0.001	0.034	0.001	0.035	0.994	0.002	0.954	0.005	1
Cox	2000	-0.002	0.002	0.110	0.002	0.116	0.716	0.010	0.961	0.004	1

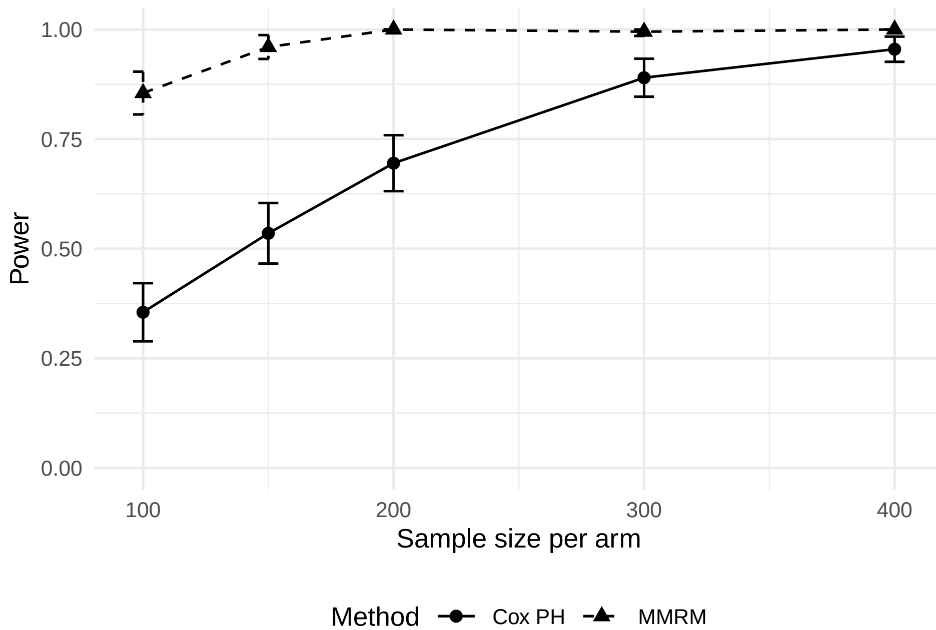

Figure 1: Empirical power as a function of per-arm sample size for MMRM and Cox proportional hazards, at 200 replicates per cell. MMRM uniformly dominates Cox across the explored range, with the gap widest at $n_{\mathrm{per\,arm}} = 100$.

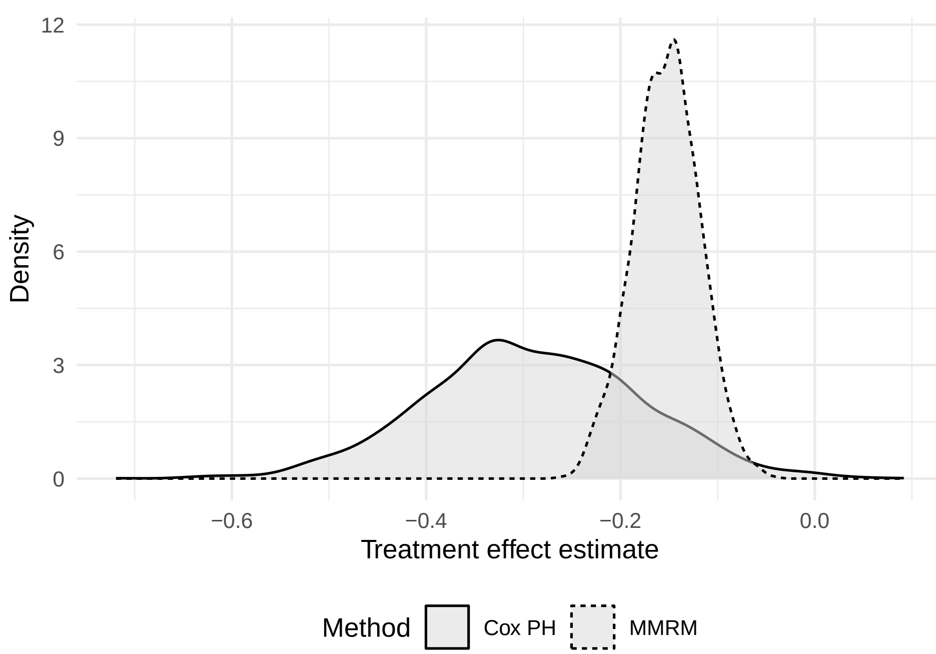

Figure 2: Distribution of per-replication treatment-effect estimates across 2,000 simulated trials at $n_{\mathrm{per\,arm}} = 200$. MMRM and Cox estimands are on different scales (continuous CDR-SB rate vs log hazard ratio); the comparison is qualitative.

3.3 Figures

4 Discussion

The historical review reveals that the convergence of AD trials on MMRM and change-score estimands resulted from three reinforcing forces: (1) the regulatory requirement for continuous cognitive and functional co-primary endpoints, established by[5]; (2) the decisive empirical evidence against LOCF as a primary analysis method, led by[9] and codified by the[14] panel; and (3) the statistical efficiency advantages of modeling repeated continuous data relative to dichotomizing into time-to-event endpoints, as demonstrated by[19] and[20].

Oncology, by contrast, arrived at a different equilibrium because its primary endpoint—death—is inherently binary and irreversible, because the proportional hazards framework has deep regulatory precedent, and because no single continuous severity measure plays the role that ADAS-Cog or CDR-SB plays in AD[1,2].

We expect the simulation results to confirm the theoretical predictions from the literature: MMRM should yield substantially greater power than Cox regression when both are applied to CDR data from the same simulated trial, because the continuous analysis uses information from the full trajectory of scores at every visit, whereas the survival analysis reduces this trajectory to a single binary event. The power advantage should be most pronounced in the MCI-to-mild-dementia transition modeled here, where conversion rates are modest and many participants in a 24-month trial will not reach the CDR ≥ 1.0 threshold, resulting in heavy right-censoring that limits the information available to the Cox model.

Several caveats warrant emphasis. First, the comparison is inherently asymmetric: the MMRM uses more information from each participant (all visits) than the Cox model (only the first conversion event). Second, the survival endpoint is limited by the visit schedule; true conversion may occur between visits, introducing interval censoring that the standard Cox model does not accommodate. Third, the relative efficiency depends on the threshold chosen; a threshold that most participants cross during follow-up would shift the balance toward the survival analysis. Fourth, the clinical relevance of the two estimands differs: a treatment effect expressed as slowing of CDR-SB decline addresses a different clinical question than one expressed as delay to a diagnostic milestone. Both may be meaningful to patients and regulators, and the choice between them should be driven by the estimand framework of[15] rather than by statistical convenience alone.

5 Future research

1. **Additional covariance structures.** The present simulation assumes MAR dropout. Extensions under MNAR mechanisms (e.g., pattern-mixture models, selection models) would illuminate the robustness of each method under more adversarial missingness.

2. **Nonlinear trajectories.** AD progression is not strictly linear, particularly in the transition from MCI to dementia. Simulation of sigmoidal or piecewise trajectories would test the sensitivity of MMRM (which makes no linearity assumption) relative to

slope-based alternatives.

3. **Alternative survival formulations.** Interval-censored survival models and multi-state models that accommodate transitions among CDR stages ($0.5 \to 1 \to 2 \to 3$) could recover some of the information lost by standard Cox regression.
4. **Joint modeling.** Joint models of longitudinal CDR trajectories and the time-to-conversion event could unify the two analytic approaches, potentially offering both the efficiency of longitudinal modeling and the clinical interpretability of a time-to-event estimand.
5. **Software tools.** An R package implementing the simulation framework and providing power calculation utilities for both MMRM and survival designs in AD trials would facilitate trial planning.
6. **Applied validation.** Retrospective application of both methods to completed trial datasets (e.g., from the ADCS or ADNI) would ground-truth the simulation findings in real data with genuine missingness patterns and measurement properties.
7. **Estimand alignment.** Further work is needed to clarify how the ICH E9(R1) estimand framework should be applied when the same underlying data support both change-score and time-to-event estimands, particularly regarding the handling of intercurrent events such as treatment switching or rescue medication.

6 References

6.1 Morris et al. (2019) ADEMP Compliance

We audited this simulation study against the reporting standards proposed by Morris, White, and Crowther[23]. The full audit is recorded at `docs/morris-audit-2026-04-17.md`.

Verdict: Not compliant.

Key gaps identified:

- Most report chunks are `eval=FALSE`; results are never produced by the rendered report.
- Estimand extractor at `sim_study.R:92` falls back to `grep('trt')[1]`, which can silently pick a `trt:visit_f` interaction term.
- No Monte Carlo SE on any metric.

A remediation plan is recorded in the audit file and will be taken up in a subsequent revision.

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